

# Initial-data convergence is redundant in the scale-separated coarse-graining theorem

FA/Banach run packet for arXiv:2602.17579

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## Source conjecture

The source is B. Hilder, P. van Meurs, and U. Sharma, *Non-equilibrium functional inequalities for finite Markov chains*, arXiv:2602.17579. Its Theorem 5.6 concerns a finite scale-separated Markov chain on  $\mathcal{X} = \mathcal{Y} \times \mathcal{Z}$ , with generator

$$L^\varepsilon = \varepsilon^{-1}Q + G,$$

coarse marginal  $\widehat{\mu}_t^\varepsilon$ , and effective law  $\eta_t^\varepsilon$ . Among its hypotheses is the convergence, as  $\varepsilon \rightarrow 0$ , of the microscopic initial law  $\mu_0^\varepsilon$  and the two coarse initial laws  $\widehat{\mu}_0^\varepsilon, \eta_0^\varepsilon$ .

Remark 5.7 says that convergence of  $\mu_0^\varepsilon$  is used only to obtain a uniform bound on  $\mathcal{H}(\mu_0^\varepsilon \mid \rho^\varepsilon)$ , where  $\rho^\varepsilon$  is the stationary law of  $L^\varepsilon$ , and conjectures that this assumption can be removed.

## Result and proof intuition

We prove more: the entire convergence bullet is unnecessary. The two uses of convergence in the cited proof have direct uniform replacements.

1. Since the finite-state stationary laws  $\rho^\varepsilon$  converge to a strictly positive law, their smallest coordinates are uniformly positive. The elementary estimate

$$\mathcal{H}(\mu \mid \rho^\varepsilon) \leq \log \frac{1}{\min_x \rho^\varepsilon(x)}$$

is then uniform over *all* probability laws  $\mu$ .

2. Uniform positivity of the two coarse initial laws propagates on every fixed time interval. Indeed, cancellation of the fast within-cluster generator gives a scalar inequality  $p'_y(t) \geq -Rp_y(t)$ , where  $R$  depends only on the bounded inter-cluster rates.

**Theorem 1.** *In Theorem 5.6 of arXiv:2602.17579, delete the hypothesis that*

$$\mu_0^\varepsilon, \quad \widehat{\mu}_0^\varepsilon, \quad \eta_0^\varepsilon$$

*converge as  $\varepsilon \rightarrow 0$ . Retain all the other hypotheses, in particular uniform strict positivity of  $\widehat{\mu}_0^\varepsilon, \eta_0^\varepsilon$  and the initial  $O(\varepsilon)$  coarse relative-entropy estimate. Then the conclusions (60) and (61) of that theorem remain valid, with constants independent of sufficiently small  $\varepsilon$ .*

## Proof

We use the notation and assumptions of Theorem 5.6. The state spaces are finite. Hilder–Sharma, *Quantitative Coarse-Graining of Markov Chains*, arXiv:2201.10256, Lemma 4.3, proves

$$\rho^\varepsilon \longrightarrow \rho \in \mathcal{P}_+(\mathcal{X}) \quad (\varepsilon \rightarrow 0).$$

Consequently, after decreasing  $\varepsilon_0$  if necessary, there is  $r > 0$  such that

$$\min_{x \in \mathcal{X}} \rho^\varepsilon(x) \geq r \quad (0 < \varepsilon < \varepsilon_0). \quad (1)$$

For every  $\mu \in \mathcal{P}(\mathcal{X})$ , with  $0 \log 0 = 0$ ,

$$\begin{aligned} \mathcal{H}(\mu \mid \rho^\varepsilon) &= \sum_{x \in \mathcal{X}} \mu(x) \log \frac{\mu(x)}{\rho^\varepsilon(x)} \\ &\leq \sum_{x \in \mathcal{X}} \mu(x) \log \frac{1}{\rho^\varepsilon(x)} \leq \log \frac{1}{r}. \end{aligned} \quad (2)$$

Here we used  $\mu(x) \leq 1$ , hence  $\mu(x) \log \mu(x) \leq 0$ . This gives exactly the uniform microscopic entropy bound for which convergence of  $\mu_0^\varepsilon$  is invoked in Remark 5.7 and in the last paragraph of the proof of Hilder–Sharma Theorem 4.5.

We next replace the use of convergence of the two coarse initial laws. For  $y \in \mathcal{Y}$ , let

$$R_y = \max_{z \in \mathcal{Z}} \sum_{z' \in \mathcal{Z}} G_{y,1-y}(z, z'), \quad R = \max_{y \in \mathcal{Y}} R_y.$$

These numbers are finite and independent of  $\varepsilon$ . Summing the microscopic forward equation over the fiber  $\Lambda_y = \{y\} \times \mathcal{Z}$ , all terms from  $\varepsilon^{-1} Q_y$  cancel because  $Q_y$  is a Markov generator. The remaining inflow from the other fiber is nonnegative, while the outflow is at most  $R_y \widehat{\mu}_t^\varepsilon(y)$ . Thus

$$\frac{d}{dt} \widehat{\mu}_t^\varepsilon(y) \geq -R_y \widehat{\mu}_t^\varepsilon(y). \quad (3)$$

The effective off-diagonal rate is an average of the same microscopic inter-fiber rates:

$$N^\varepsilon(y, 1-y) = \sum_{z, z' \in \mathcal{Z}} \rho^\varepsilon(z \mid y) G_{y,1-y}(z, z') \leq R_y.$$

The effective forward equation therefore similarly gives

$$\frac{d}{dt} \eta_t^\varepsilon(y) \geq -R_y \eta_t^\varepsilon(y). \quad (4)$$

The retained first bullet of Theorem 5.6 supplies a constant  $c_0 > 0$  such that both initial coarse laws are at least  $c_0$ . Gronwall’s inequality applied to (3) and (4) yields, for every  $T < \infty$ ,

$$\inf_{\substack{0 \leq t \leq T \\ y \in \mathcal{Y} \\ 0 < \varepsilon < \varepsilon_0}} \{\widehat{\mu}_t^\varepsilon(y), \eta_t^\varepsilon(y)\} \geq c_0 e^{-RT} > 0. \quad (5)$$

This is precisely the finite-time,  $\varepsilon$ -uniform positivity used in the proof of Hilder–Sharma Theorem 4.5 to control the logarithm. The cited proof originally obtains it from convergence of the coarse initial laws to averaged dynamics; (5) supplies it directly.

We have now replaced every use of the deleted convergence bullet. The source paper already removes the formerly assumed uniform log-Sobolev constant by continuity of its generalized log-Sobolev constant. Applying the rest of the proof of Theorem 5.6 unchanged gives the finite-time  $O(\varepsilon)$  estimate. In that estimate the entropy dissipation is bounded by its initial value, which

is uniformly bounded by (2). The source’s long-time generalized log-Sobolev argument then combines with the finite-time estimate exactly as written to give

$$\mathcal{H}_{\eta_t^\varepsilon} \left( \frac{\widehat{\mu}_t^\varepsilon}{\eta_t^\varepsilon} \right) \leq C_1 \varepsilon e^{-\beta t} \quad (t \geq 0),$$

and hence its time-uniform consequence. This proves Theorem 1.  $\square$

## Verification, scope, and novelty

The source conjecture and convergence bullet were checked on PDF page 25. The exact use of microscopic initial convergence was checked on page 21 of the supporting paper: it is invoked only to bound  $\mathcal{H}(\mu_0^\varepsilon \mid \rho^\varepsilon)$ . Its Lemma 4.3 proves the finite-state convergence to a positive stationary law used in (1). The fiber summation leading to (3) is also the cancellation used throughout both papers’ scale-separated setup.

The result retains uniform strict positivity at time zero. Removing that condition while keeping an estimate from  $t = 0$  is different: the logarithm used in the relative-entropy comparison is singular at zero. The cited paper does give delayed-time estimates on  $[\delta, T]$  without this positivity, but no all-time strengthening is claimed here.

Before promotion, the run indexes were searched for arXiv:2602.17579, its title, Theorem 5.6, and initial-data convergence phrases. A bounded arXiv/web search on 13 August 2026 used the exact conjecture sentence and close variants. It found the source and arXiv:2201.10256, but no later answer.

## References

- B. Hilder, P. van Meurs, and U. Sharma, *Non-equilibrium functional inequalities for finite Markov chains*, arXiv:2602.17579.
- B. Hilder and U. Sharma, *Quantitative Coarse-Graining of Markov Chains*, SIAM J. Math. Anal. 56 (2024), 913–954; arXiv:2201.10256.

**Human-review recommendation.** Review as a short, likely valid full solution with strengthening. The key checks are (2), cancellation of  $Q_y$  in (3), and the uniform effective-rate bound.